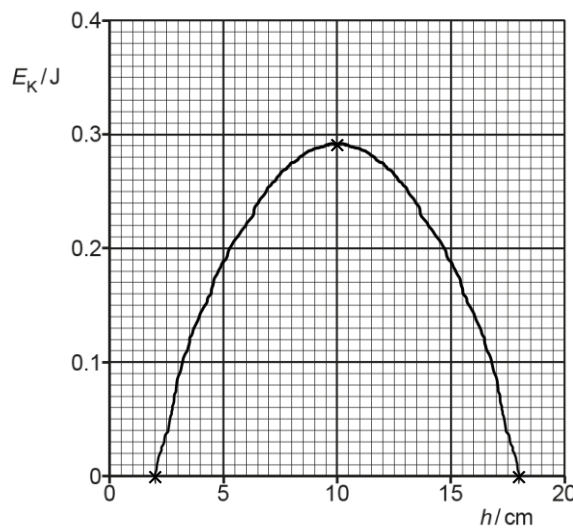


|  |                                                                                                                                                                                                                                                                                                                                   |  |    |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----|
|  | <p>At distance <math>R = 4r</math>, <math>v = \sqrt{\frac{2GM}{4r}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 6.0 \times 10^{24}}{4 \times 6.4 \times 10^6}} = 5.59 \times 10^3 \text{ m s}^{-1}</math></p> <p>Change in speed <math>= 6.46 \times 10^3 - 5.59 \times 10^3 = 8.7 \times 10^2 \text{ m s}^{-1}</math></p> |  | A1 |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----|

|      |                                                                                                                                                                                                                                                                                                                                                                                                                   |   |                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|
| 4a   | $a = -\omega^2 x$<br>$a$ = acceleration, $x$ = displacement <u>from equilibrium position</u> and $\omega$ = angular frequency                                                                                                                                                                                                                                                                                     | E | B1             |
| 4bi  | $\omega = 2\pi / T$<br>$= 2\pi / 4.0$<br>$= 1.57$<br>$= 1.6 \text{ rad s}^{-1}$                                                                                                                                                                                                                                                                                                                                   | A | B1<br>A0       |
| 4bii | $E = \frac{1}{2}m\omega^2 x_0^2$ Or $E = \max E_k = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}m\omega^2 x_0^2$<br>$= \frac{1}{2} \times 36 \times 1.6^2 \times 0.080^2$<br>$= 0.29 \text{ J}$                                                                                                                                                                                                                           | A | B1<br>C1<br>A1 |
| 4c   | <p><u>dome-shaped curve</u>, starting and ending at <math>E_K = 0</math><br/>           maximum <math>E_K</math> shown as 0.29 J, position of peak shown at <math>h = 10.0 \text{ cm}</math><br/>           line intercepts <math>h</math>-axis at <math>h = 2.0 \text{ cm}</math> and at <math>h = 18.0 \text{ cm}</math></p>  | D | B1<br>B1<br>B1 |

|    |                                                                                   |   |      |
|----|-----------------------------------------------------------------------------------|---|------|
| 5a | The field strength at a point equals the negative of the potential gradient there | A | (B2) |
|----|-----------------------------------------------------------------------------------|---|------|